

B.E 4th Semester(CST) Final Examination 2015
Indian Institute of Engineering Science and Technology Shibpur,
Howrah-711103, India.
Discrete Structures CS-401

Time: 3 hrs.

Full Marks: 70

Answer any three questions from each half. Two marks are reserved in each half for neatness. Use separate answer script for each half.

1st Half

1. (a) Let R_1 and R_2 be symmetric relations. Is $R_1 \cap R_2$ also symmetric? Is $R_1 \cup R_2$ also symmetric?
(b) Let S be the set of subroutines of a computer program. Define the relation ρ by $P\rho Q$ if subroutine P calls subroutine Q during its execution.
 - (i) Describe the transitive closure of ρ .
 - (ii) For which subroutines P does (P, P) belong to the transitive closure of ρ ?
 - (iii) Describe the reflexive closure of the transitive closure of ρ .

[5+6]
2. (a) Suppose that $(S, \preceq_1)$ and $(T, \preceq_2)$ are posets. Show that $(S \times T, \preceq)$ is a poset where $(s, t) \preceq (u, v)$ if and only if $s \preceq_1 u$ and $t \preceq_2 v$.
(b) Draw the Hasse diagram for divisibility on the set.
 - (i) $\{1, 2, 3, 4, 5, 6, 7, 8\}$.
 - (ii) $\{1, 2, 3, 6, 12, 24, 36, 48\}$.

[4+4+3]
3. (a) Show that the set of all computer programs in a particular programming language is countable.
(b) Show that if S is a set, then there does not exist an onto function f from S to $\mathcal{P}(S)$, the power set of S .
(c) Prove that the cardinality of the set of real numbers $\mathbb{R}$ is greater than the cardinality of the set of integer numbers $\mathbb{Z}$.

[3+4+4]
4. (a) What do you understand by linear recurrence relation? Solve following two recurrences
 - (i) $f_n = f_{n-1} + f_{n-2}$ for $n > 1$ and $f_0 = 0, f_1 = 1$.
 - (ii) $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$.

[1+3+3+4]

(b) Find out the generating function for the sequence $\langle 2, 5, 10, 17, \dots \rangle$.

SECOND HALF

5. a) What do you mean by logical consequence?

Define logical consequence in propositional logic.

State two theorems on logical consequence on Propositional logic and prove them.

b) Determine whether the following argument is valid.

If 2 is a prime, then it is the least prime. If 1 is a prime, then 2 is not the least prime.

The number 1 is not a prime. Therefore, 2 is the least prime.

$[(1+1+5)+4]$

6. a) Define Prenex Normal Form.

Find Prenex Normal Form of the formula:

$\exists x (P(x) \rightarrow \neg \exists y (P(y) \rightarrow (Q(x) \rightarrow Q(y)))) \wedge \forall x (P(x) \leftrightarrow \forall y Q(z))$

b) Consider the following formula:

$\exists x P(x) \rightarrow \forall x P(x)$

i) Prove that this formula is always true if the domain D contains only one element.

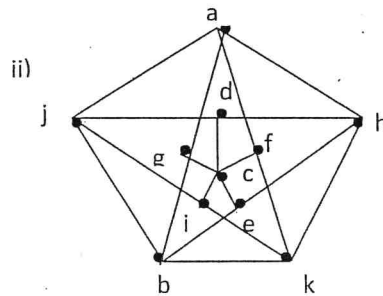
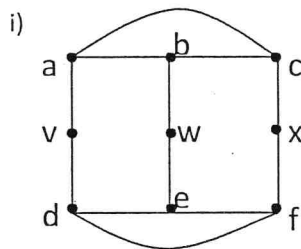
ii) Let $D=\{a,b\}$. Find an interpretation over D in which A is evaluated to F.

c) Some patients like all doctors. No patient like any quack. Prove that no doctor is a quack.

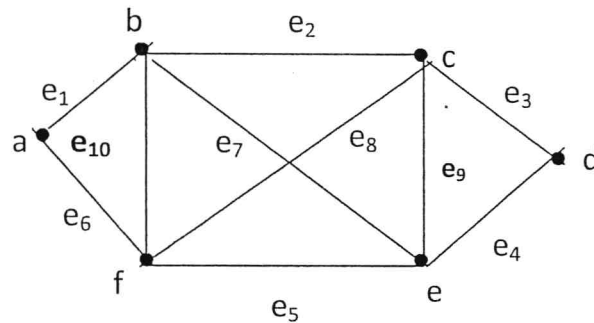
$[(1+2)+(1.5 \times 2)+5]$

7. a) For the following, either construct a Hamiltonian cycle in the given graph,

or prove that the graph is not Hamiltonian.



b)) Use Fleury's algorithm to construct Euler's tour for the following graph.



[(4+4)+3]

8. a) Justify Proof by Contradiction with appropriate truth tables.

Use proof by contradiction to establish that for any integer n , $n^2 + 2$ is not divisible by 4.

b) Justify Proof by Contra positive with appropriate truth tables.

Use proof by contra positive to establish that for any integer n , if n^2 is odd then n is odd.

c) What is proof by cases? Justify proof by cases using truth tables.

Let A, B and C be three inhabitants of an island where there are either "knights" or "knaves". The "knights" always tell the truth and "knaves" never tell the truth. Two Individuals are of the same type if they are either both knights or both knaves. Suppose A says, "B is a knave" and B says "A and C are of the same type". Use proof by cases to prove that C is a knave.

[3+3+5]